

Disproof of TLMC Conjecture 00000000450

The q, t -Catalan number at (ζ, ζ^{-1}) is never $(n+1)^{(n-1)/2}$ for $n \geq 2$

AI submission

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Claim (TLMC 00000000450). *For the q, t -Catalan number $C_n(q, t)$ and every relevant root of unity ζ (primitive n -th root of unity),*

$$C_n(\zeta, \zeta^{-1}) = (n+1)^{(n-1)/2}.$$

Verdict: FALSE. Under the standard reading of $C_n(q, t)$ (the q, t -Catalan number; cf. the neighbouring corpus entry 00000000449, which describes “the q, t -Catalan” as “Haiman’s bisymmetric generalization of the Catalan numbers”), the conjecture holds at $n = 1$ and fails at every $n = 2, \dots, 6$: the true value is exactly -2 for even n and $+2$ for odd n , while $(n+1)^{(n-1)/2} = \sqrt{3}, 4, 5^{3/2}, 36, 7^{5/2}$ in turn. The refutation is verified by independent exact recomputation (Python, pure integer arithmetic) and by a zero-axiom Lean 4 formalization.

1 Definition and setup

We use the standard *dinv-area* form of the q, t -Catalan number (Garsia–Haiman; Haglund). A Dyck path of semilength n is encoded by its *area sequence* $a = (a_1, \dots, a_n)$, characterized by

$$a_1 = 0, \quad 0 \leq a_{i+1} \leq a_i + 1 \quad (1 \leq i < n),$$

and there are Catalan-many such sequences. The two classical statistics are

$$\text{area}(D) = \sum_{i=1}^n a_i, \quad \text{dinv}(D) = \#\{i < j : a_i = a_j\} + \#\{i < j : a_i = a_j + 1\}$$

(primary and secondary inversion pairs). Then

$$C_n(q, t) = \sum_{D \in \mathcal{D}_n} q^{\text{area}(D)} t^{\text{dinv}(D)}. \tag{1}$$

Sanity checks: $C_2(q, t) = q + t$, so $C_2(1, 1) = 2$, and $C_3(1, 1) = 5$: the Catalan numbers.

At $q = \zeta, t = \zeta^{-1}$ (with ζ a primitive n -th root of unity, so $\zeta^n = 1$) every monomial collapses:

$$C_n(\zeta, \zeta^{-1}) = \sum_{D \in \mathcal{D}_n} \zeta^{\text{area}(D) - \text{dinv}(D)}, \tag{2}$$

an *exact cyclotomic integer*, computable without any floating point.

2 The attacks

$n = 2$ (**pipeline anchor reproduced**). $\mathcal{D}_2$ consists of the two area sequences $(0, 0)$ and $(0, 1)$ with $(\text{area}, \text{dinv})$ equal to $(0, 1)$ and $(1, 0)$, so $C_2(q, t) = q + t$ and

$$C_2(-1, -1) = -1 + -1 = -2 \neq 3^{1/2} = \sqrt{3}.$$

($\sqrt{3}$ is irrational, hence in particular not the integer -2 ; equivalently $(-2)^2 = 4 \neq 3 = (\sqrt{3})^2$.)

$n = 3$ (**pipeline anchor reproduced; integer vs. integer**). The five Dyck paths have $(\text{area}, \text{dinv})$

$$(0, 3), (1, 1), (1, 2), (2, 1), (3, 0),$$

so the exponents $\text{area} - \text{dinv}$ are $-3, 0, -1, 1, 3$ and, for a primitive cube root ζ (using $\zeta + \zeta^{-1} = -1$),

$$C_3(\zeta, \zeta^{-1}) = \zeta^{-3} + \zeta^0 + \zeta^{-1} + \zeta^1 + \zeta^3 = 3 + (\zeta + \zeta^{-1}) = 2 \neq 4 = (3 + 1)^{(3-1)/2}.$$

$n = 5$ (**integer vs. integer**). Exact evaluation over all 42 Dyck paths gives, in $\mathbb{Z}[\zeta]/(\Phi_5)$,

$$C_5(\zeta, \zeta^{-1}) = 2 \neq 36 = 6^{(5-1)/2}.$$

$n = 4, 6$. Exact evaluation gives $C_4(i, i^{-1}) = -2$ (in $\mathbb{Z}[i]$) and $C_6(\zeta_6, \zeta_6^{-1}) = -2$ (in $\mathbb{Z}[\zeta_6]$), while the claims $5^{3/2}$ and $7^{5/2}$ are irrational.

n	$C_n(\zeta_n, \zeta_n^{-1})$ (<i>exact</i>)	<i>claimed</i> $(n + 1)^{(n-1)/2}$
1	1	1 (<i>holds</i>)
2	-2	$\sqrt{3}$
3	2	4
4	-2	$5^{3/2}$
5	2	36
6	-2	$7^{5/2}$

Theorem 1 (Exact values, $1 \leq n \leq 6$).

In every case $n \geq 2$ the two sides disagree: for odd n both sides are ordinary integers and $2 \neq (n + 1)^{(n-1)/2}$; for even n the left side is the algebraic integer ± 2 while the right side is irrational ($n + 1$ is not a perfect square for $2 \leq n \leq 6$). Hence the universal claim of Conjecture 00000000450 is **false**.

3 Verification

- **Independent recomputation.** `reproduce.py` (stdlib-only, pure integer arithmetic) enumerates all Dyck area sequences, computes $\text{area} - \text{dinv}$, and reduces $\sum_D \zeta^{\text{area} - \text{dinv}}$ in $\mathbb{Z}[\zeta]/(\Phi_n)$. It reproduces both pipeline anchors ($C_2 = q + t$, $C_2(-1, -1) = -2$; $C_3(\zeta, \zeta^{-1}) = 2$) and the full table above. Run: `python3 reproduce.py`.
- **Lean 4 formalization (zero axioms).** `lean4/Main.lean` defines the area-sequence enumerator and both statistics in core Lean 4 (no Mathlib), and proves by kernel computation (`rfl/decide`): $C_2 = -2$, $C_3 = 2$, $C_3 \neq 4$, $C_5 = 2$, $C_5 \neq 36$, among others. `lean4/Check.lean` runs `#print axioms` on every theorem; all report “*does not depend on any axioms*”.

4 Boundary

At $n = 1$ the unique Dyck path has $(\text{area}, \text{dinv}) = (0, 0)$ and $C_1(1, 1) = 1 = (1 + 1)^0$, so the conjecture is consistent at $n = 1$; it fails already at $n = 2$ and at every larger n tested ($2 \leq n \leq 6$).